Creating a GitHub Action from Scratch

DevOps practices use automation to carry out repetitive tasks, leaving you free to focus on the more challenging but fun parts of building software. This article will explain how to use GitHub Actions, the automation platform baked into all GitHub projects for free.

GitHub Actions makes it easy to automate all the software workflows that you see in your program — from code reviews, branch management and issue triaging, to even a complete CI/CD pipeline.

GitHub Actions runs workflows when other events happen in your repository, such as when code is pushed, a pull request is opened or an issue is created. 'Workflows' are YAML specs that are defined within the *.github/workflows* directory in your repository. These workflows contain one or more 'jobs' that can be run sequentially or in parallel. Each job will run inside its own virtual machine 'runner' or inside a container. Each job has one or more steps that either run a script you define, or run an 'action' which is a reusable extension that can simplify your workflow. These actions can be either those that you create yourself, or use for free from the GitHub Marketplace. Figure 1 illustrates how all of this works together in accomplishing the automation that you need.

Let's understand this better through an example that many of us might have seen in our CI pipelines — when code is pushed, create a build (Figure 2).

In this example, the 'event' that happens in your repository is a 'push' event, and you need automation to create a build of your code. To accomplish this, you need to create a workflow YAML file within the *.github/workflows* directory, and define all the necessary components of jobs, runners and steps necessary for your build to take place.

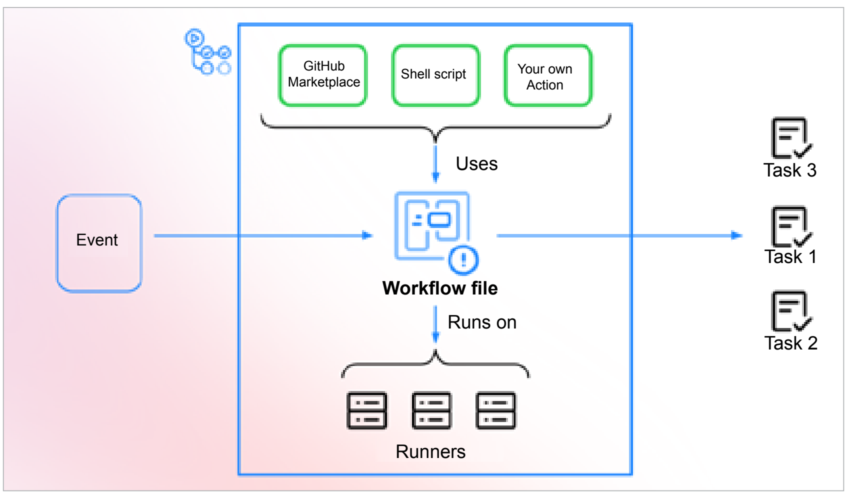

Figure 1: How GitHub Actions work

Workflow files

Let's look at how to write workflows in YAML to define the automation that you need. Figure 3 gives an example of what a workflow file will look like, along with all the elements necessary.

In this example, you can see the 'on' key that defines the event in your repository. On clicking this, the workflow is triggered to run automatically using GitHub Actions. Your workflow can have multiple triggers, and even add qualifiers to